How people talk about financial markets on Bluesky

Ege Aydemir · Web and Social Media Search and Analysis, A.Y. 2025/26 · Project type 1: Social Media Analysis (network + content)

1. Objective

The project started from a simple question: when people talk about "the market" on social media, who is actually being listened to, and is the mood of that conversation somethng we could use? Since many journalists and economists moved from X to Bluesky, I chose Bluesky as the platform and looked at one month of English posts about financial markets, from 13 August to 12 September 2026.

I split the topic into three smaller conversations, which I call streams, so that I always had something to compare against: **equities** (the stock market and the S&P 500), **macro** (the Fed, its new chair Kevin Warsh, and the FOMC) and **crypto** (bitcoin and ethereum). A number like "clustering is 0.03" means little on its own, but it starts to mean something next to the same number for a neighboring stream.

Finally, I wanted to answer four things:

1. What does the reply network look like: a small group of accounts everyone replies to, or a conversation spread across many people?
2. Does the conversation split into communities, and do those communities follow my three streams or something else?
3. What is the tone of each stream and each community, which people and institutions attract the most negativity, and do people reply to others who feel the same way?
4. Does the volume or the mood of the conversation move together with the market itself (the S&P 500 and Bitcoin)?

The practical angle I had in mind is a "social listening" team at a broker or asset manager that wants to know which accounts and groups to keep an eye on, and how seriously to take the platform's mood.

2. Data

2.1 How I collected it

I used the official Bluesky API (the `atproto` Python library) with an app password created for this project, and I only read public posts. The collector pages through the `Searchs Posts endpoint 100 posts at a time inside a fixed time window, waits a bit between requests, and if the API says I am going too` fast it sleeps until the limit resets. So the collection is not a sample of popular posts; it is everything that matched my queries inside the window.

Before the real collection I did a pilot: twelve candidate search terms on the same three days, comparing posts per day, the share of posts that were replies (a sign that people actually talk, not just broadcast) and the number of distinct authors, plus a random sample of posts read by eye. This changed my query list a lot. `Nasdaq` had plenty of posts but almost no replies, because one account was posting automated "Trading Halted / Resumed" alerts; `"Wall Street"` was big but mostly off-topic; `crypto` on its own was huge and mostly promotion; `Powell` was the wrong name, because Kevin Warsh has been Fed chair since May 2026. In the end I kept seven queries: `"stock market"` and `"S&P 500"` for equities; `"the Fed"`, `Warsh` and `FOMC` for macro; `bitcoin` and `ethereum` for crypto.

The search only gives me the posts that contain my keywords, not the conversations under them. For the network I need to know who replied to whom, so for every collected post with at least one reply (8,908

posts) I fetched the whole thread with `getPostThread` and walked it recursively, saving each reply together with the author of the post it answers. This took about two hours because of the rate limit.

2.2 What the data looks like

The seven queriesthat are returned 34,533 unique posts; crypto is about half of them, equities and macro are similar in size. The thread fetching added 28,556 replies, so in total I have text for about 60,000 posts from roughly 20,000 accounts.

Not all of that is usable. Since financial topics attract a lot of bots, we have to be carefull.: price tickers, whale alerts, the same promotional text posted hundreds of times. Instead of removing accounts by hand I wrote four simple rules and flagged an account if it broke any of them: more than 15 posts a day on average, half or more of its posts matching a bot-like template ("SIGNAL", "Take Profit", "Market Cap:"), only broadcasting , or half or more of its posts being duplicates of a text that appears at least five times. The rules flagged 73 accounts, which sounds like nothing, but they wrote 18.6% of all the posts. Removing them barely changed the reply network (about a hundred nodes out of sixteen thousand), which confirms that these accounts talk to nobody. One more surprise: in wrestling slang "the Fed" means WWE, so the macro stream also contained a few hundred posts about wrestling, which I flagged with a keyword list and dropped together with their threads.

After cleaning, the network has 15,973 users and 18,980 reply links, and the text corpus (human, English, non-duplicate posts) has 44,958 posts from 18,627 authors.

3. Models

3.1 The network

Each user is a nodes and each reply is a directed edge from the person replying to the author of the post being replied to, weighted by the number of replies. Replies to yourself are dropped, because they are just people continuing their own thread. I built one graph for the whole conversation and one per stream.

To describe the shape of the network I used the standard measures from the course (density, reciprocity, giant component, clustering, average path length) and compared them with a random graph (Erdős–Rényi) with the same number of nodes and edges. That is the small-world test: much higher clustering than the random graph, with similar or shorter distances. To find who matters I computed in-degree (how many people reply to you), out-degree (how many people you reply to), PageRank, betweenness (are you on the path between groups) and closeness, and checked with a rank correlation whether they pick the same people. For communities I ran Louvain and three alternatives (greedy modularity, label propagation, FluidC) on the giant component and compared them by modularity; since Louvain has a random start, I also ran it with five seeds and measured the agreement with NMI, to be sure the communities are in the data and not in the seed.

3.2 The content

Preprocessing is done with spaCy in one pass: remove links and handles, lemmatise, drop stop words and my own query words (otherwise "fed" and "bitcoin" would dominate every word cloud), and extract named entities (people, organisations, places) at the same time.

For sentiment I used three methods on purpose: two lexicon-based ones (VADER and AFINN) and a transformer trained on tweets (`cardiffnlp/twitter-roberta-base-sentiment-latest`). I wanted to see how much they disagree and pick one with a reason. The reason came from a validation on 2,000 human-labelled tweets from the tweet_eval benchmark: RoBERTa got 72% accuracy, VADER and AFINN stayed around 52–55%. So RoBERTa is the main label in the results and VADER is kept as a second opinion and as a continuous score. For emotions I used the NRC lexicon (eight basic emotions such as fear, anger, trust, anticipation). The named entities were mapped to a few normalised names ("the Federal Reserve", "the Fed" and "Fed" become one thing), and for each entity I computed the share of positive and negative posts that mention it.

3.3 Putting the two together

Notebook 5 joins the two. Each post gets the Louvain community of its author, so every community can be described by its most replied-to account, its distinctive words, its top entities and its mood. For the echo-chamber question I checked whether linked users have a similar average tone (assortativity) and whether a reply carries the same polarity as the post it answers, compared with chance. For the market question I downloaded daily prices of SPY and BTC-USD with `yfinance` and correlated daily post counts and negativity with returns and volume, same day and one day ahead.

4. Results

4.1 A small world with a few very big hubs

The network is very sparse: an average user is connected to about two other people, and roughly a quarter of the reply links go both ways. Still, 70% of all users sit in one giant connected component, and inside it the average distance between two people is about 6 steps, while the random graph of the same size needs about 10. Clustering is around 80 times higher than in the random graph. Short paths plus local groups: the conversation is a small world, exactly the pattern the course describes for social networks.

The degree distribution explains the short paths. Most people receive zero or one reply, and a handful of accounts receive hundreds. The most replied-to account, atrupar.com (an independent political journalist), got replies from 738 different people, and the other names at the top (meidastouch.com, muellershewrote.com, carlquintanilla, kevinmkruse) are political commentators, news accounts, historians and one CNBC reporter. I looked up their profiles to be sure: they are not financial analysts. On Bluesky, the finance conversation is led by political and news accounts.

The people who send the most replies are a completely different group; the rank correlation between in-degree and out-degree is basically zero. In-degree and PageRank rank almost the same people (0.97), so PageRank added little. Betweenness was the interesting one: besides the big hubs it highlights a few accounts with only two or three connections that sit between two clusters, so many of the shortest paths between groups pass through them. These are the brokers, and they never appear in the hub list.

4.2 Communities follow people, not topics

Louvain found 67 communities with a modularity of 0.86, which is very high, and greedy modularity got almost the same score. Five different seeds agree at NMI 0.90, so the structure is stable, and only 12.5% of the edges connect different communities.

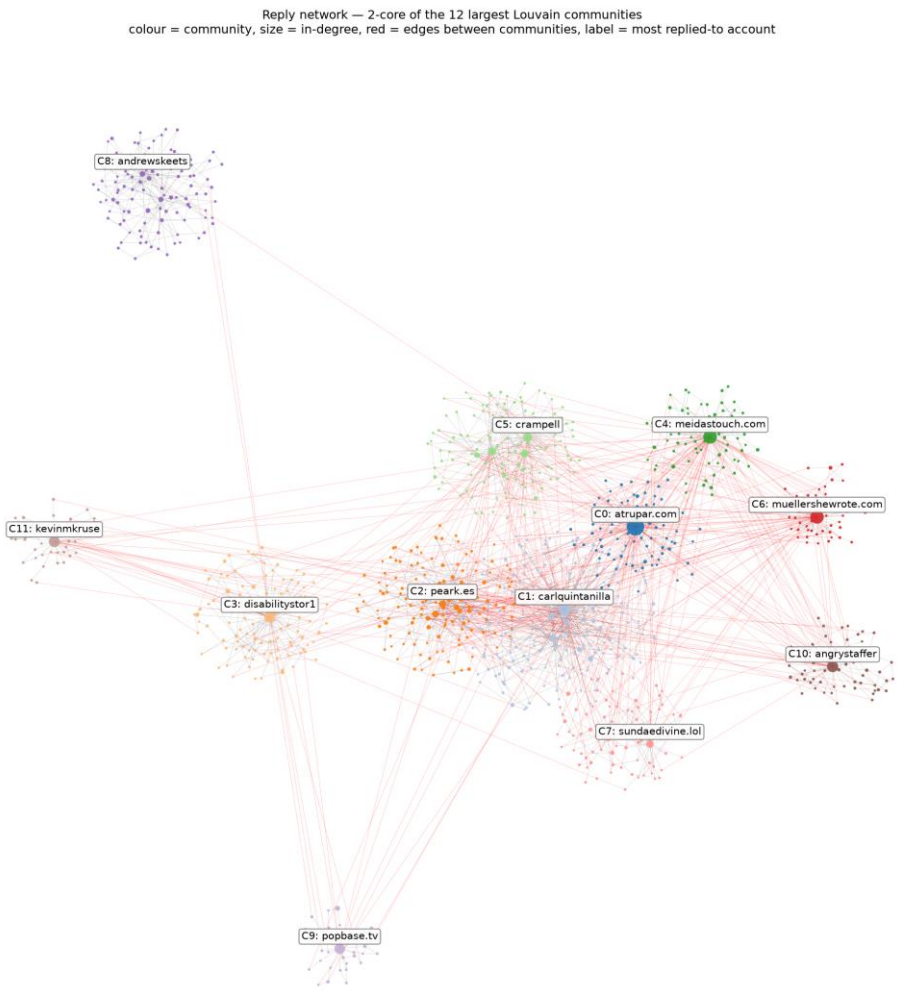


Figure 1. The 2-core of the twelve largest communities. Colour is the community, node size is the number of people who replied to that account, red lines are the links between communities. Each community is labelled with its most replied-to account.

The figure shows what the communities actually are: each one is a star around one big account, and the label of every cluster is a person, not a topic. When I cross-tabulated the communities with my three streams, the largest group in eleven of the twelve biggest communities was "reply-only", people who never used my keywords themselves but replied to somebody who did. The assortativity by stream is 0.31, positive but moderate: people prefer to reply within their own subject, but the streams are not closed rooms. Equities and macro talk happen inside the same audiences; crypto is a bit more separate and also the least conversational stream (smallest giant component).

Some communities are not about finance at all. C8 is a leftover of the wrestling problem, C9 (popbase.tv) is about an AI-generated Bitcoin film with Gal Gadot, and C3 and C11 are academic threads that happened to hang from one keyword post. That is how keyword collection works: one matching post drags its whole thread in.

4.3 Equities are the angriest stream, crypto the happiest

With RoBERTa, 45% of the equities posts are negative and 13% positive. Macro is 40% negative and only 6% positive, with the majority neutral, which fits the reporting style of Fed news. Crypto is the only stream where positive and negative are balanced (about 20% each). The word lists explain part of it: in equities the top words are "trump", "economy", "crash"; in macro "rate", "inflation", "hike"; in crypto "price", "blockchain", "buy". Replies fetched from threads are more negative than the original posts (46% vs 34%), so the tone gets harsher as a discussion goes deeper.

VADER agrees on the big picture (crypto is the least negative) but is far more optimistic: it calls 40% of the corpus positive where RoBERTa says 13%, and the two agree on only about half of the posts. Given the

validation result I trust RoBERTa more, but the disagreement is a result in itself: sarcasm and financial jargon ("crash", "short", "bear") are not handled well by a word list.

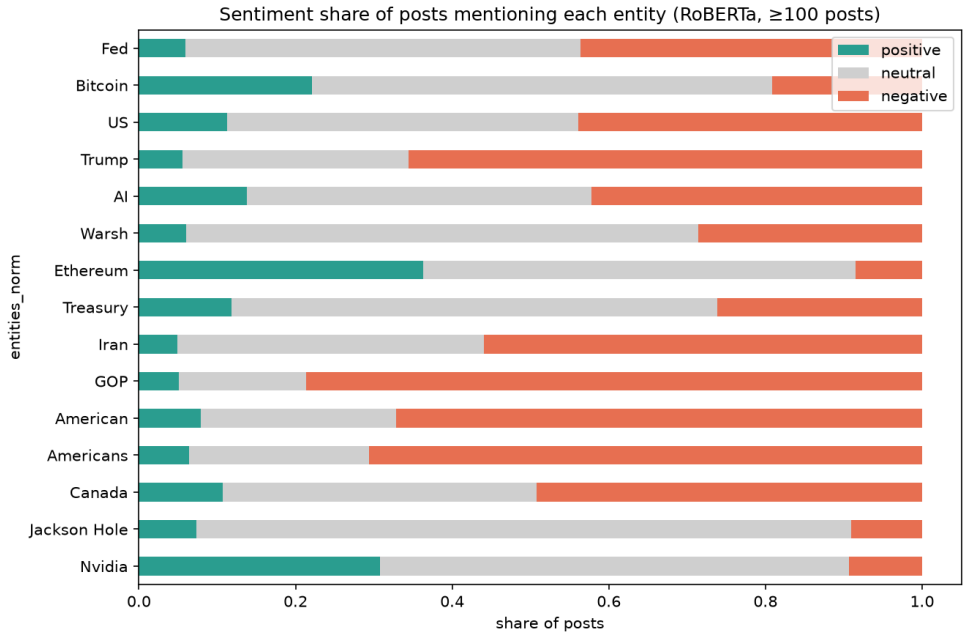


Figure 2. Share of positive and negative posts among the posts that mention each entity (RoBERTa, entities with at least 100 posts).

The entity view shows where the negativity goes. Posts mentioning the Fed are 44% negative and 6% positive, posts about Trump 66% negative, the GOP 79%. Kevin Warsh is discussed mostly neutrally, and Jackson Hole is almost pure reporting. On the other side, Bitcoin, Ethereum and Nvidia are the only entities with more positive than negative posts. So the negativity of the finance conversation on Bluesky is mostly political negativity attached to the economy, not fear about specific assets.

The community profiles say the same thing from the network side. The communities around market reporters and economists (carlquintanilla, peark.es, crampell) are the calmest, around 37–42% negative, with distinctive words like "PCE", "productivity", "VIX", "Nixon", "appoint" (a discussion about political pressure on Fed chairs through history). The communities around political accounts (muellershewrote, angrystaffer, meidastouch) are 63–69% negative and talk about jobs, votes and the government. The NRC emotions add one detail: the political communities score highest on fear and anger, while among the finance communities the one around carlquintanilla scores highest on anticipation, the language of "what happens next". The mood depends on which community you are in more than on the news of the day.

4.4 Do people reply to like-minded people?

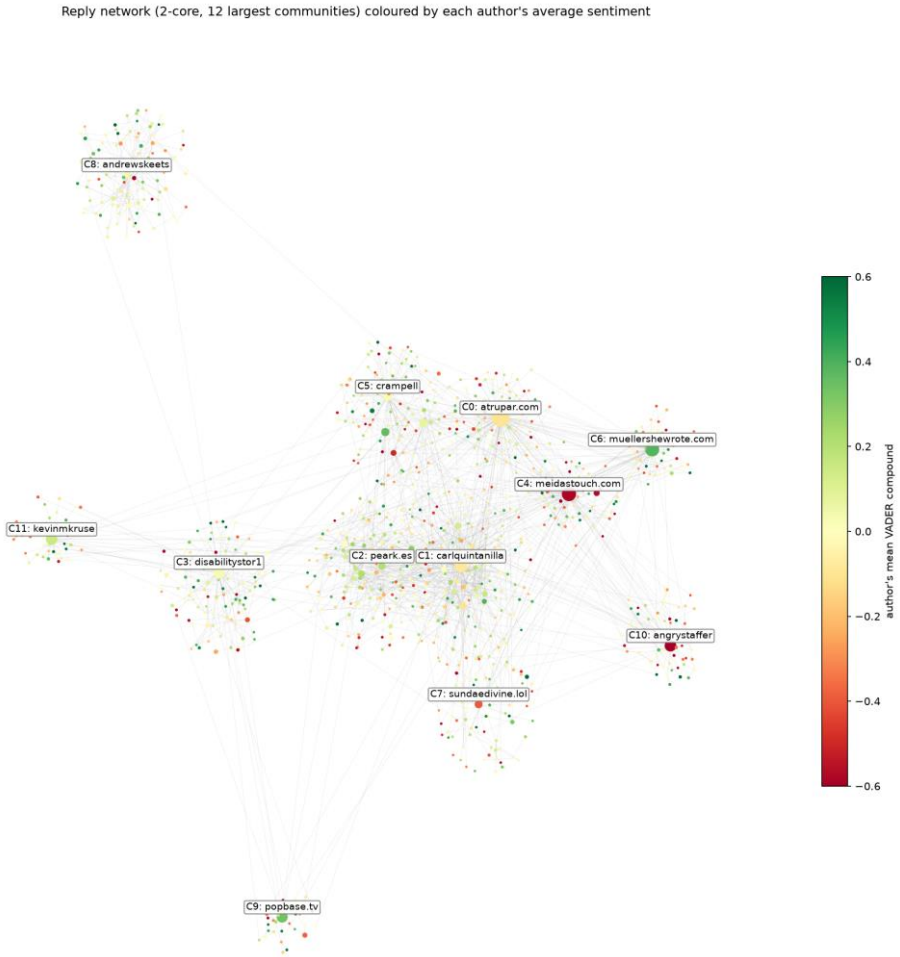


Figure 3. Same layout as Figure 1, but each author is coloured by the average VADER score of their own posts (green positive, red negative).

Not much, but negativity is contagious. At the author level, the assortativity of average sentiment is 0.04, essentially zero, and the figure shows it: every community mixes green, yellow and red dots, nobody sorts into positive and negative camps. It also shows that a hub's own tone is not the tone of its audience: muellershewrote is green (positive wording in the account's own posts) while its community is the most negative of all.

At the reply level there is a clear effect, though. A reply has the same polarity as the post it answers 51% of the time, against 40% expected by chance, and the effect is one-sided: a negative post gets a negative reply 56% of the time, a positive post gets a positive reply only 28% of the time. So there is no echo chamber in the sense of separate camps, but negativity propagates down threads much more than positivity does.

4.5 The conversation and the market

The timeline is where the streams behave most differently. The macro stream is quiet most of the month and then explodes on two days, the Jackson Hole symposium (Warsh's first big speech as chair, 28 August) and the jobs report (4 September), each with about a thousand posts. I checked the distinctive words of those days to be sure they are really the events and not something else, and they are ("jacksonhole", "keynote", "kevinwarsh" on the first; "jobs", "slash", "halt" on the second). Macro negativity is interesting too: it drops to its lowest point right at Jackson Hole, because that day is dominated by neutral reporting of the speech, and then climbs to its peak around the jobs report. Equities negativity stays high all month and does not react much to the events. Crypto has its own calendar: its biggest day is 1 September, and the distinctive words of that day are "gadot", "movie", "gal", the AI film story, not a market event at all.

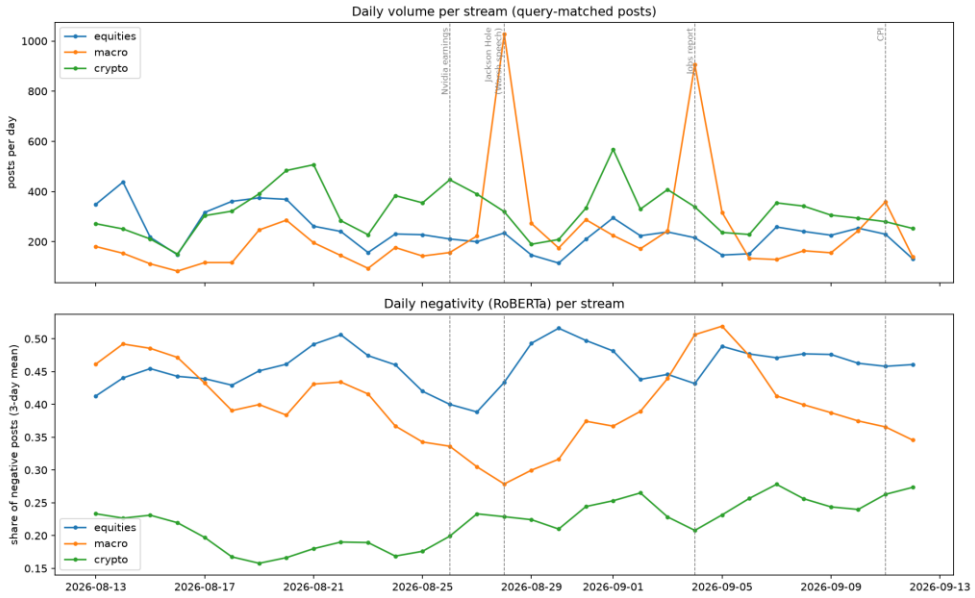


Figure 4. Daily volume (top) and daily share of negative posts (bottom) per stream, with the four scheduled market events of the month.

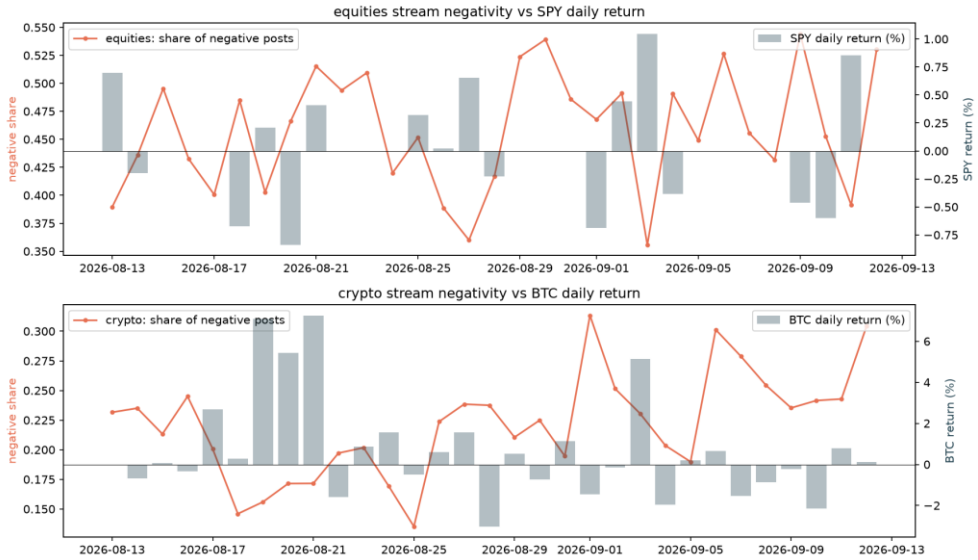


Figure 5. Daily negativity of the equities and crypto streams against the daily return of SPY and Bitcoin.

The last step is exploratory and I want to be careful with it, because one month gives only 17 trading days for SPY and 31 days for Bitcoin. For Bitcoin the picture is clear: the number of crypto posts is strongly correlated with same-day trading volume (r = 0.70) and with the size of the price move, and the share of negative posts goes up when the price goes down (r = −0.42). For equities, the only significant relation is that negativity is higher on days when SPY falls (r = −0.59). The one-day-ahead correlations are weaker and inconsistent, so I read all of this as "the conversation reacts to the market", not "the conversation predicts the market".

5. Limitations

A few things I would do differently with more time. The collection depends on keywords, so a post about the market that does not use my seven terms is invisible, and a thread that mentions "the Fed" once drags in unrelated conversations (wrestling, films, universities). My bot rules are simple thresholds; they catch the obvious broadcasters but surely miss some accounts and probably flag a few very active humans. Sentiment was validated on general tweets, not on financial posts, and even the best model was wrong on

more than a quarter of them, so the percentages above are tendencies, not exact figures. The market comparison covers one month and says nothing about causality. And betweenness was computed on a sample of source nodes to keep the running time reasonable.

6. Conclusion

In a few sentences: the finance conversation on Bluesky is a small world organised around a dozen very large accounts, and those accounts are political commentators and news reporters rather than finance people. The communities are their audiences, and the same audience talks about stocks and about the Fed, so the topic split I imposed at collection time is not how the network organises itself. The tone is mostly negative, with the negativity attached to politics and the economy in general, while the assets themselves (Bitcoin, Ethereum, Nvidia) are discussed positively. People do not sort into positive and negative camps, but negative posts breed negative replies. Volume spikes on scheduled macro events and, for crypto, tracks the market's daily moves, without evidence that it predicts them.

For the social-listening team I imagined at the start: monitor the dozen hub accounts and the reporter communities around them if you want an informative signal, treat the political communities as a mood indicator rather than a market signal, use a transformer rather than a word list for tone, and watch the volume around scheduled events rather than the raw level of negativity, which on this platform is high all the time.